

Problem Statement: Enterprise Knowledge Assistant Using RAG

Overview

Organizations store large volumes of information across PDFs, SOPs, technical documents, knowledge bases, and internal portals. As the volume of data grows, employees spend significant time searching for relevant information, leading to reduced productivity and repeated dependency on subject matter experts.

Traditional keyword-based search systems often fail to understand the intent behind user queries and may not retrieve the most relevant information. Additionally, standalone LLMs cannot reliably answer organization-specific questions because they do not have access to internal enterprise knowledge.

Objective

Build a **Retrieval-Augmented Generation (RAG)** solution that enables users to ask questions in natural language and receive accurate, context-aware answers generated from enterprise documents.

The solution should:

- Ingest enterprise documents from multiple sources.
- Perform semantic search to retrieve relevant information.
- Generate grounded responses using retrieved content.
- Reduce hallucinations by providing document context to the LLM.
- Scale as organizational knowledge grows.

Technical Stack

Framework

- **LangChain**
 - Document ingestion
 - Text chunking
 - Retrieval pipeline orchestration
 - Prompt management

Embedding Model

- **Open-Source Embedding Model**

- Examples: BGE-M3, E5-Large, Nomic Embed
- Converts documents and user queries into vector embeddings for semantic search.

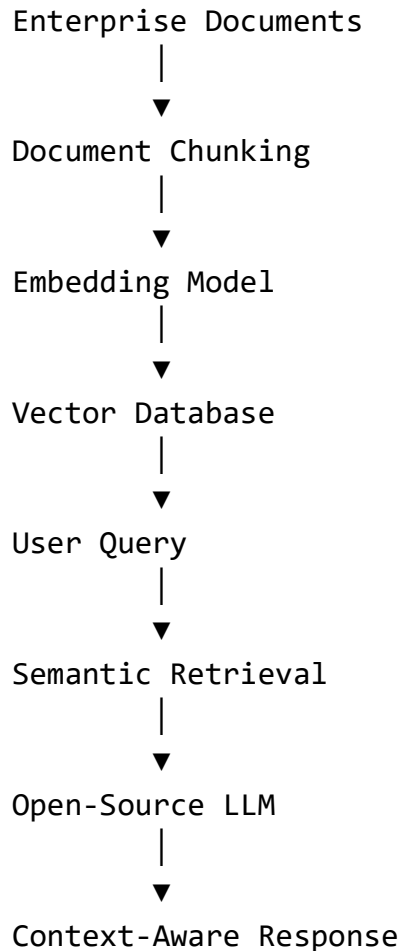
Large Language Model (LLM)

- **Open-Source LLM**
 - Examples: Llama 3, Qwen, Mistral, DeepSeek
 - Generates responses using retrieved document context.

Vector Database

- Stores document embeddings and enables similarity-based retrieval.

High-Level Workflow



Expected Outcome

A conversational AI assistant capable of providing accurate, source-grounded answers from enterprise knowledge using **LangChain**, **open-source embedding models**, and

open-source LLMs, while maintaining data privacy and reducing reliance on manual document searches.